Abhirath **Anand**

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Education

University of Michigan *Ann Arbor, USA*

MS in Bioinformatics, **GPA**: 3.98/4 *Aug. 2024 - Present*

Working with **Dr. Evan Snitkin** for developing methods to study infectious disease transmission, specifically in the context of antimicrobial resistant pathogens in healthcare settings.

Birla Institute of Technology and Science, Pilani (BITS Pilani) *Pilani, Rajasthan, India*

BE in Computer Science, **GPA**: 7.85/10 *Nov. 2020 - Jun. 2024*

Thesis: Worked with **Dr. Vinti Agarwal** in collaboration with **Dr. Ben Woodcroft**, Center for Microbiome Research (CMR), Queensland University of Technology on prediction models for salinity in Earth's microbial communities.

Experience

Research Experience

Summer Intern *Memphis, Tennessee, USA*

University of Tennessee Health Science Center (UTHSC) *Jun. 2025 - Aug. 2025*

- Selected as part of the Biomedical Data Science Internship (BDSI) program in the Division of Biostatistics under the Department of Preventive Medicine.
- Worked with **Dr. Saunak Sen** and **Dr. Gregory Farage** on developing a Julia package designed to streamline working with metabolomics data.
- Developed separate packages for preprocessing, visualization and statistical analysis, further integrating them under a single umbrella package to serve as one-stop solution for metabolomics matrix data.
- Presented my work at a poster session community science exhibition held in Rhodes College, Memphis at the end of the internship.

Summer Research Intern *Mumbai, Maharashtra, India*

Tata Institute of Fundamental Research (TIFR) *May 2023 - Jul. 2023*

- Selected as part of the Visiting Students' Research Program (VSRP) in the Department of Biological Sciences (DBS) at TIFR.
- Worked with **Dr. Amitesh Anand** on computational insights into the bioenergetics of microbes.
- Analysed Complex I of the electron transport chain in microbes, with a focus on its evolution and its spread across the microbial tree of life.

Undergraduate Research Assistant *Remote*

Microbial Interactions Lab, University of Southampton *Oct. 2022 - Dec. 2023*

- Worked with **Dr. Franklin Nobrega** on computational insights related to genomic plasticity in *Salmonella*.
- Focussed on using computational tools to analyse how genomic plasticity could impact factors like pathogenicity, antibiotic resistance, metal resistance and antiphage defence systems.

Undergraduate Research Projects *Pilani, Rajasthan, India*

BITS Pilani *2022 - 2023*

- Worked with **Dr. Balakumaran Chandrasekar** on analysing proteomics data using computational approaches, with a focus on phylogenetic orthology inference and protein-protein interaction modelling during the time period Aug. 2023 - Dec. 2023.
- Worked with **Dr. Vinti Agarwal** on graph machine learning algorithms for text classification during the time period Nov. 2022 - May 2023.
- Worked with **Dr. Tejasvi Alladi** in exploring machine learning applications in network security and in detecting unmanned aerial vehicles during the time period Jan. 2022 - Dec. 2022.

Engineering Experience

Google Summer of Code Contributor *Remote*

Metalhead.jl, FluxML *May 2022 - Sep. 2022*

- Worked on improving the deep learning for computer vision ecosystem in the Julia programming language, specifically in the FluxML ecosystem for deep learning with a focus on the Metalhead.jl library.
- Refactored the Metalhead.jl library to incorporate more modern training techniques and newer-age models.
- Developed a general purpose API for models to make them hackable and allow machine learning researchers and engineers to start experimentation easily.

Summer Intern *Remote*

GERMI (Gujarat Energy Research and Management Institute) *May 2022 - Jul. 2022*

- Worked on extreme learning machines (ELMs) and their uses.
- Explored using ELMs in tandem with autoencoders for online condition monitoring systems in industrial environments.

Publications

Hidden Markov Model-Based Prokaryotic Genome Space Mining Reveals the Widespread Pervasiveness of Complex I and Its Potential Evolutionary Scheme

Akshay Shirsath, Snehal V Khairnar, **Abhirath Anand**, Divya M Prabhakaran, and Amitesh Anand

Genome Biology and Evolution, 2025. **DOI**: *10.1093/gbe/evaf154*

Comprehensive blueprint of *Salmonella* genomic plasticity identifies hotspots for pathogenicity genes

Simran Krishnakant Kushwaha, Yi Wu, Hugo Leonardo Avila, **Abhirath Anand**, Thomas Sicheritz-Pontén, Andrew Millard, Sandhya Amol Marathe, and Franklin L. Nobrega

PLOS Biology, 2024. **DOI**: *10.1371/journal.pbio.3002746*

Invited Talks

Datageddon, APOGEE *Pilani, Rajasthan, India*

Introduction to Machine Learning and Deep Learning *Mar. 2023*

Introduced first-year students to the basic concepts of machine learning and deep learning. Also led hands-on sessions involving coding, with a focus on understanding and implementing concepts like the perceptron algorithm and backpropagation.

Workshops Attended

Simons Centre for Living Systems, NCBS *Bangalore, India*

"Physics of Life" Monsoon School *19 - 28 Jun. 2023*

- Attended a 10 day "Monsoon School" organised by the **Simons Centre for Living Systems at the National Centre for Biological Sciences (NCBS)** designed to introduce students from quantitative backgrounds to interdisciplinary research in biology.
- Chosen as part of a **select cohort of 50 students** from across India.
- Attended lectures and hands-on sessions on a variety of topics in biology.
- Worked on a project to model predator-prey interactions in real time using the Viscek model.

IIIT Hyderabad *Remote*

Summer School on Artificial Intelligence *18 July 2022 - 19 Aug 2022*

- Attended a month long summer school on Artificial Intelligence organised by the Centre for Visual Information Technology at the International Institute of Information Technology, Hyderabad **(CVIT, IIIT Hyderabad)**.
- Attended lectures and hands-on sessions on a variety of topics in AI, including machine learning, deep learning, and reinforcement learning.
- Was exposed to a variety of application areas of AI, including computer vision, natural language processing, and robotics.

Projects

2023 **Compiler for toy language "ERPLAG",** GitHub repository *Compiler Design*

2022 **FluxML/Metalhead.jl,** GitHub repository *Computer Vision*

Clubs and Societies

Sally Robotics *Pilani, Rajasthan, India*

Team Lead and Computer Vision Researcher *May 2021 - Oct. 2023*

- Joined Sally Robotics as a Computer Vision Researcher working on problems related to self-driving cars and autonomous navigation in the context of Indian roads.
- Was promoted to **Team Lead** in May 2022 for the duration of the academic year 2022-23. Headed a team of 20+ members across four subsystems (Computer Vision, SLAM, Navigation and Mechanical) working on problems related to autonomous driving.
- Mentored and guided students in the team, specifically in computer vision and deep learning, and facilitated regular discussions on research papers and new developments in the field.
- Was responsible for procuring funding for the team worth ₹150k from university grants.

BITS Model United Nations Society (BITSMUN) *Pilani, Rajasthan, India*

Member *Nov. 2020 - Nov. 2021*

- As part of the Model United Nations Society, engaged in weekly discussions on international affairs.
- Participated in several Model United Nations conferences conducted by various colleges across India.

Honors & Awards

2020	**All India Rank (AIR) 1183,** Joint Entrance Examination (JEE) Mains	*India*
2020	**All India Rank (AIR) 1640,** Joint Entrance Examination (JEE) Advanced	*India*
2019	**All India Rank (AIR) 389,** Kishore Vaigyanik Protshahan Yojana (KVPY) SA	*India*
2018	**NTS Scholarship,** National Talent Search Scholarship	*India*

Skills

Programming	C, Python, Julia, Java, LaTeX
Technical Skills	Machine Learning, Deep Learning, Computer Vision, Bioinformatics
Languages	English, Hindi, Tamil